



SOFTWARE ENGINEERING 1

MediSync: An Integrated Healthcare Management System for National University Dasmariñas

Presented by:

Name	Position	Signature
De Guzman, John Kenneth L.	Leader	
Blanco, John Kenneth I.	Member	
Calitis, Mary Edvianne	Member	
Conde, Reister Gerald G.	Member	
Mosinabre, Paula Bianca P.	Member	
Vitan, Nataniel John I.	Member	

Table of Contents

A. BUSINESS REQUIREMENTS SPECIFICATION.....	1
1. INTRODUCTION.....	1
1.1 Problem.....	1
1.2 Objectives.....	1
1.3 Purpose.....	1
1.4 Definition of Terms.....	1
1.5 References.....	1
2. PROJECT DESCRIPTION.....	1
2.1 Project Perspective.....	1
2.2 Conceptual Framework.....	1
2.3 User Characteristics.....	1
2.4 Project Components.....	2
2.5 Project Constraints.....	2
2.6 User Interface.....	2
2.7 Assumptions and Dependencies.....	2
2.8 Proposed Future Requirements.....	2
3. REQUIREMENTS ENGINEERING.....	2
3.1 Requirements Elicitation.....	2
3.2 User Requirements.....	3
3.3 Functional Requirements.....	3
3.4 Non-Functional Requirements.....	3
3.5 Specific Requirements.....	3
4. FEASIBILITY STUDY.....	3
4.1 Operational Feasibility.....	3
4.2 Technical Feasibility.....	3
4.3 Economic Feasibility.....	4
4.4 Schedule Feasibility.....	4
B. TECHNICAL REQUIREMENTS SPECIFICATION.....	4
5. SYSTEM ANALYSIS.....	4
5.1 Current System Analysis.....	4
5.2 Proposed System Analysis.....	4
5.3 Process Model.....	4
6. SYSTEM DESIGN.....	4
6.1 Use Case Diagram.....	4
6.3 Context Diagram.....	4
6.4 Database Schema.....	5
6.5 Screen Layout / User Interface Design.....	5
APPENDICES.....	5



Revision History

Revision	Date	Author	Description of Revision

A. BUSINESS REQUIREMENTS SPECIFICATION**1. INTRODUCTION****1.1 Problem****Guide:**

Describe the problem or situation that the proposed system intends to address. Explain the current issues experienced by the users or organization.

1.2 Objectives**Guide:**

State the objectives of the proposed system.

Include:

- One general objective
- Specific objectives that describe what the project aims to achieve

Objectives should be clear and related to the identified problem.

1.3 Purpose**Guide:**

Explain why the system is being developed and what it is intended to accomplish. Describe the expected value or benefit of the proposed system.

1.4 Definition of Terms**Guide:**

List important technical terms, abbreviations, or concepts used in the document.

Arrange the terms alphabetically when possible.

Term Definition**1.5 References****Guide:**

List the books, websites, articles, documents, policies, or other sources used in preparing the project documentation.

Use one consistent citation format throughout the document.APA.

2. PROJECT DESCRIPTION**2.1 Project Perspective****Guide:**

Describe the project in a broader context.

Explain whether the proposed system is:

- A new system
- An improvement of an existing process
- A replacement for an existing system
- A component of a larger system

Briefly explain how the project fits into the organization or environment where it will be used.

2.2 Conceptual Framework

Guide:

Present a visual representation of how the project works at a high level.

The framework should show the relationship between important elements of the project, such as:

- Inputs
- Processes
- Outputs

Include a brief explanation of the framework after the figure.

2.3 User Characteristics

Guide:

Identify the expected users of the system.

For each user type, describe:

- Role
- Responsibilities
- Level of access
- Relevant skills or characteristics

User Type Role Main Responsibilities

2.4 Project Components

Guide:

Identify the major components or modules of the proposed system.

For each component, provide a short description of its purpose.

Component/Module Description

2.5 Project Constraints

Guide:

Identify the limitations or restrictions that may affect the development or operation of the project.

These may include:

- Time limitations
- Available resources
- Technology limitations
- Budget limitations
- Team skills
- Hardware or software requirements

2.6 User Interface

Guide:

Briefly describe how users will interact with the system.

At this stage, provide a general description of the interface. Detailed screen designs will be included under **System Design**.

Describe:

- Main pages or screens
- Navigation
- User interaction
- General usability considerations

2.7 Assumptions and Dependencies

Guide:

Assumptions are conditions believed to be true for the project to work as planned.

Dependencies are external factors, systems, tools, services, or resources that the project relies on.

Clearly list and explain each item.

2.8 Proposed Future Requirements

Guide:

List features or improvements that are not included in the current version of the project but may be considered in future iterations.

Do not include these as part of the current required system features.

3. REQUIREMENTS ENGINEERING

3.1 Requirements Elicitation

Guide:

Explain how the project requirements were gathered.

Identify the methods used to collect information from stakeholders or potential users.

Describe:

- Who participated
- What data gathering method was used
- Why the method was selected
- How the gathered information helped identify the requirements

Possible supporting documents may be included in the Appendices.

3.2 User Requirements

Guide:

Describe the needs and expectations of the users in simple and understandable language.

Focus on what the users need the system to help them accomplish.

Group the requirements according to user type when appropriate.

3.3 Functional Requirements

Guide:

List the specific functions and features that the system must perform.

Each requirement should:

- Be clear
- Be specific
- Be testable
- Describe one identifiable system function

Use the following format:

ID Functional Requirement Description

FR-01

FR-02

FR-03

Use a consistent numbering format throughout the document.

3.4 Non-Functional Requirements

Guide:

Describe the quality attributes and performance expectations of the system.

These requirements may relate to:

- Performance
- Security
- Usability
- Reliability

- Compatibility
- Availability

Use the following format:

ID Non-Functional Requirement Description

NFR-01

NFR-02

NFR-03

3.5 Specific Requirements

Guide:

This section provides a more detailed description of the major requirements and features of the proposed system.

Organize the specific requirements according to the main modules, features, or functions of the system.

4. FEASIBILITY STUDY

4.1 Operational Feasibility

4.2 Technical Feasibility

4.3 Economic Feasibility

4.4 Schedule Feasibility

B. TECHNICAL REQUIREMENTS SPECIFICATION

5. SYSTEM ANALYSIS

5.1 Current System Analysis

Guide:

Describe how the current process or existing system works.

Explain the steps involved in the current process and identify the problems, limitations, or inefficiencies.

A flowchart or process diagram may be included.

5.2 Proposed System Analysis

Guide:

Describe how the proposed system will address the problems identified in the current system.

Explain:

- What processes will be improved
 - What tasks will be automated or simplified
 - How users will interact with the proposed system
 - How the proposed system differs from the current process
-

5.3 Process Model

Guide:

Create a visual representation of a major process in the proposed system.

Use an appropriate process modeling technique discussed in class.

The diagram should clearly show:

- Start and end points
- Major activities
- Decisions, when applicable
- Flow of information or actions

Include a short explanation of the diagram.

6. SYSTEM DESIGN

Note: Only the required diagrams and designs below should be included. Students are not required to create additional UML diagrams unless instructed.

6.1 Use Case Diagram

Guide:

Create a Use Case Diagram that identifies:

- System users or actors
- Major system functions
- Relationships between actors and use cases

The diagram should focus only on the major functions identified in the Functional Requirements.

6.2 Context Diagram

Guide:

Create a Context Diagram that shows the proposed system as a single process and identifies the external entities that interact with it.

The Context Diagram provides a high-level view of the system and shows:

- The proposed system
 - External entities or users that interact with the system
 - Information or data entering the system
 - Information or data produced by the system
-

6.3 Database Schema

Guide:

Present the proposed database design.

Include:

- Main entities or tables
- Important attributes
- Primary keys
- Foreign keys, when applicable
- Relationships between entities

An ERD or database schema diagram may be used.

Keep the design focused on the database requirements of the proposed system.

6.4 Screen Layout / User Interface Design

Guide:

Present the proposed design of the major system screens.

Include only the important screens necessary to demonstrate the main system features.

For each screen, provide:

- Screen name
- Screen name
- Brief description of the main components

Screens may be designed using an appropriate prototyping or design tool.

APPENDICES